BRENDA
The Comprehensive Enzyme Information System

Information on EC 3.2.1.8 - endo-1,4-beta-xylanase

TEMPERATURE STABILITY	ORGANISM	UNIPROT	COMMENTARY	LITERATURE
0 - 45	Pomacea insularus	-	pH 6.9, 10 min, stable	171584
0 - 50	Penicillium citrinum	-	the retention of 50% of its maximum activity is observed at 0°C, retains 65% of its activity after 1-4 h incubation at 50°C	681711
10	Sphingobacterium sp. TN19	D8L2Y2	purified recombinant enzyme, pH 6.5, 40% remaining activity	707250
100 ⊟	Bacillus licheniformis	-	partially purified extracellular xylanase, inactivation after 120 min	752334
	bacterium enrichment culture clone Xyl8B8	H9ZGD1	30 min, 28% residual activity	730872
	Halalkalibacterium halodurans	-	3 h, no loss of activity	731022
	Lentinula edodes	-	1 h, purified Xyl16-3, 50% remaining activity	655228
	Saccharolobus solfataricus	-	half-life of 47 min at 100°C	679721
	Thermoclostridium stercorarium	-	pH 7.0, 10 min, about 40% loss of activity. pH 5.5, 10 min, complete inactivation	171657
		-	10 min, retains 50% of its activity at pH 6.1, retains 58% of its activity at pH 6.4, completly inactivated at pH 5.8	171661
	Thermomonospora sp.	-	half-life 15 min	654413
	Thermomyces dupontii	-	half-life 60 min	741629
		-	half-life 1 h	741630
	[Caldibacillus] cellulovorans	Q9L8L8	15% remaining activity of recombinant N-terminal + catalytic domains	656718
110	Aspergillus fumigatus	-	60% of XylF2 activity is retained at 110°C	682991
20	Sphingobacterium sp. TN19	D8L2Y2	purified recombinant enzyme, pH 6.5, 65% remaining activity	707250
20 - 50	Demequina sp. JK4	B9VSZ3	purified recombinant enzyme, stable	709491
20 - 60	Lentinula edodes	-	more than 50% of the optimal xylanase activity is retained when the temperature is increased from 20 to 60°C after 240 min, at 40°C the xylanase maintains 93% of the optimal activity after a 240 min reaction, the enzyme retains 78.12% and 41.64% of the maximum activity at 50 and 60°C, respectively. at temperatures up to 60°C, xylanase activity decreases sharply with time	681351
20 - 70 ⊟	Acidothermus cellulolyticus	A0LR95	purified recombinant enzyme, 30 min, completely stable at 20-50°C, 90% activity is retained at 60°C, and 70% at 70°C, inactivation at 80°C	750497
	Thermobifida alba	-	1 h, stable	171636
20 - 80	Acidothermus cellulolyticus	A0LR95	the enzyme remains completely stable from 20 to 50°C, while about 90 and 70% of the activity is retained at 60 and 70°C after 30 min. The melting temperature is 80°C	750497

25	Euphausia superba	-	stable up to for at least 10 min, rapid inactivation above, isozymes A and B	655257
30 ⊟	Bacillus sp. (in: firmicutes)	-	stable for up to 72 h	667012
	Cladosporium sp.	A0A346PXS9	3 h, 50% loss of activity	750679
		A0A346PXS9	purified recombinant enzyme, 3 h, loss of 50% activity	750679
	Cutaneotrichosporon cutaneum	-	7 days, 55% loss of activity	171625
	Penicillium griseofulvum	-	stable at 30°C for 60 min, with more than 95% of the activity remaining	677745
		-	purified recombinant PgXynB is stable at 30 °C, pH 4.5 for 240 min, with more than 85% of the activity remaining	707567
	Talaromyces funiculosus	Q9HFH0	stable at 30°C for 60 min, with more than 95% of the activity remaining	677745
	Thermochaetoides thermophila	Q8J1V6	purified recombinant enzyme expressed from gene with intron, half-life is 1959 min	709894
	uncultured microorganism	E2DQY5	xylan as substrate, remains stable after treatment for 1 h	717134
30 - 40	Bacillus cereus	-	the enzyme displays thermostability at 30°C and 40°C with 48.56% and 45.97% remaining activity after 180 min, respectively. At 50 °C, the enzyme loses about 46% of its activity after 40 min	770012
30 - 45	Aspergillus tubingensis	-	3 h, completely stable	707209
30 - 50 ⊟	Aspergillus flavus	B8NJ86	purified native, stable at	750988
		B8NJ86	purified native, 30% activity remaining	750988
	Bacillus sp. (in: firmicutes)	-	purified enzyme, highly stable at	717091
	Sphingobacterium sp. TN19	D8L2Y2	purified recombinant enzyme, pH 6.5, over 80% remaining activity, above 50°C the enzyme activity decreases rapidly after 10 min preincubation	707250
	uncultured microorganism	E2DQY5	purified recombinant RuCelA, 1 h, stable	717134
30 - 55	Neocallimastix patriciarum	A8TGA1	at temperatures below 30°C and above 55°C, enzyme activity is less than 50% of that at the optimal temperature	695791
30 - 70	Streptomyces thermocyaneoviolaceus	Q9RMM4	purified enzyme, 1 h, 65% activity remaining at 70°C, completely stable at 30-60°C, overview	707809
35 ⊟	Cladosporium sp.	A0A346PXS9	1 h, 50% loss of activity	750679
		A0A346PXS9	purified recombinant enzyme, 1 h, loss of 50% activity	750679
	Cutaneotrichosporon cutaneum	-	pH 5.0, 10 min, stable at or below	171625
	Termitomyces clypeatus	-	1 h, stable	171630
	Trametes hirsuta	-	10 min, stable	171629
35 - 55	Thermothelomyces thermophilus	-	-	742765
37 ⊟	Massilia sp. RBM26	A0A0S2I9J6	1 h, more than 96% residual activity	738866
	Streptomyces lividans	-	pH 6.0, 24 h, stable	171606

37 - 50	Bacillus pumilus	-	all isozymic forms show 100% activity at 37°C after 6 h. After 3 h, isoforms X-I, X-II, X-III, X-IV, and X-V retain nearly 86, 85, 85, 82, and 80% residual activity at 45°C and activity is reduced to 38-42% at 50°C	769914
4 - 50	Streptomyces halstedii	-	1 h, pH 6.3, stable	171639
40 ⊟	Aspergillus luchuensis	P33557	mutant T64S, 30 min, 77% resiudal activity	739002
	Aspergillus niger	-	15 min, significant inactivation	171633
		-	purified native enzyme, pH 7.0, 120 min, 80% activity remaining	710686
	Bacillus licheniformis	-	X-I and X-II are 100% stable after 1 h at 40°C	677623
	Bacillus pumilus	-	pH 5.0 or pH 8.5, 30 min, enzyme retains more than 85% of its activity	171624
	Bacillus sp. (in: firmicutes)	-	pH 6.0, 10 min, enzyme component I from strain W1, stable up to	171595
	Cladosporium sp.	A0A346PXS9	20-30 min, complete inactivation	750679
		A0A346PXS9	purified recombinant enzyme, 20-30 min, complete inactivation	750679
	Duganella sp. PAMC 27433	-	the half-life of the enzyme at 40°C is approximately 30 min	769515
	Luteimicrobium xylanilyticum	A0A286LNP2	C-terminal truncated mutant, loss of activity above 40°C	745089
	Malbranchea pulchella	-	1 h, xylanase A is stable, 25% loss of activity of xylanase B	171587
	Naganishia albida	-	5 min, stable	171631
	Penicillium citrinum	Q2PGY1	stable up to	665763
		B1B533	remains stable up to 40°C, loses its activity at 80°C	699042
		Q2PGY1	purified recombinant extracellular deglycosylated enzyme, 6 h, over 80% activity remains	752115
	Pseudopedobacter saltans	F0S6Y3	purified recombinant enzyme, pH 6.0, 90 min, stable up to	752084
	Schizophyllum commune	-	stable up to	664785
	Streptomyces olivaceoviridis	-	1 h, inactivation above	171659
	Streptomyces sp.	F8WSY7	stable up to	729010
	Talaromyces purpureogenus	A0A2H4RAC5	purified recombinant enzyme, pH 6.5, 24 h, completely stable	749635
	Termitomyces clypeatus	-	4 h, 65% loss of activity	171630
	Thermochaetoides thermophila	Q8J1V6	purified recombinant enzyme expressed from gene with intron, half-life is 959 min	709894
	Trichoderma harzianum	-	1 h, 20000 Da enzyme form is stable	171613
	Trichoderma inhamatum	-	2.2 h at 40°C and 2 min at 50°C	695668
	Trichoderma reesei	-	20 h, 10% loss of activity of xylanase I, xylanase II is stable	136301
	uncultured microorganism	E2DQY5	xylan as substrate, remains stable after treatment for 1 h	717134
40 - 50	Pediococcus acidilactici	-	purified extracellular enzyme, 2 h, over 80% activity remaining	750814

40 - 60 ⊟	Evansstolkia leycettana	-	at 60 C, the enzyme gradually loses activity, exhibiting about half activity after 3 h. At 50 C, the enzyme is fully stable for 2 h and then the activity starts to decline. At 40 C, the enzyme retains activity for 3 h, but upon a longer incubation, the activity is reduced	774449
	Marinimicrobium sp. LS-A18	A0A411LPM1	the enzyme activity does not change after incubation at 40°C for 2 h, however, it significantly decreases after incubation at 50°C or 60°C. The enzyme retains 14.8% and 2.8% of its activity after incubation at 50°C and 60°C for 30 min, respectively	768777
	Trichoderma sp.	-	purified isozyme EX1, stable at temperatures below 40°C, retains 50% activity after incubation at 50°C for 30 h, but it loses its activity rapidly at 60°C for 1 h	707762
43	Streptomyces lividans	-	pH 6.0, 24 h, about 20% loss of activity	171606
45 ⊟	Aspergillus niger	I3QKR9	1 h, the wild-type enzyme almost completely loses its activity	751243
	Aspergillus ochraceus	-	37.5% loss of activity after 90 min, stable up to 30 min	171654
	Aureobasidium melanogenum	-	the enzyme shows a half-life of 169 h at 45 °C and pH 4.5. Under these conditions and in the presence of Ca2+ (10 mM) the enzyme is even more stable (half-life of 231 h)	776226
	Bacillus pumilus	Q5EFR9	purified recombinant enzyme, 2 h, stable	717410
	Cellulomonas uda	-	1 h, 5% loss of activity	136300
	Dickeya chrysanthemi	-	instable above, loss of secondary structure and activity	655834
	Euphausia superba	-	complete loss of activity within 5 min, isozymes A and B	655257
	Penicillium citrinum	Q2PGY1	purified recombinant extracellular deglycosylated enzyme, 70% activity remains after 30 min, 17% after 3 h, and 6% after 5 h	752115
	Penicillium occitanis (nom. inval.)	-	purified enzyme, 1 h, 91% activity remaining	718338
	Penicillium sclerotiorum	-	half-life of both isozymes is about 210 min	707211
	Pomacea maculata	Q7Z1V6	recombinant EGXA and wild-type EGX lose activity quickly when Cl- is at a low concentration. In the presence of 50 mM Cl-, the rtEGXA⍁s activity loss is only 15% after 5 h incubation at 45°C. Wild-type EGX loses activity at a faster speed than the recombinant EGXA, in both 0.5 mM Cl- and 50 mM Cl-. Cellulose-binding domain increases thermal stability of the recombinant enzyme	695295
	Pseudopedobacter saltans	F0S6Y3	purified recombinant enzyme, pH 6.0, 90 min, loss of 50% activity	752084
	Talaromyces funiculosus	Q8J0K5	stable for 60 min	664181
	Thermochaetoides thermophila	Q8J1V6	purified recombinant enzyme expressed from gene with intron, 10 min, 100% activity remaining	709894
	Trametes hirsuta	-	30 min, stable up to	171629
	Trichoderma longibrachiatum	-	pH 5.0, 4.5 h, 50% loss of activity	171647
	Trichoderma viride	-	rapid loss of activity above	171623
45 - 50	uncultured bacterium	-	the enzyme remains stable when subjected to heat treatments of 45 and 50 °C for up to 1 h	775836
45 - 55 ⊟	Alternaria mali	-	slight decrease in xylanase activity between 45°C (93.4%) and 55°C (97.7%). Enzyme activity is rapidly reduced to	695683

			59.3% when the temperature is at 60°C, and there is negligible activity (5.4%) above 80°C	
	Kitasatospora sp.	A0A3T0YV67, A0A3T0YV60	purified recombinant enzyme, pH 5.0, stable for 120 min	750084
45 - 60	Nesterenkonia xinjiangensis	D1KJJ7	purified enzyme, pH 7.0, 60 min, 80% activity remaining	707216
48	Aspergillus niger	-	pH 5.5-5.6, half-life: 40 h	171598
5 - 25	Cladosporium sp.	A0A346PXS9	purified recombinant enzyme, 3 h, over 90% activity remaining	750679
50 ⊟	Acrophialophora nainiana	-	half-life 120 h	655620
	Albifimbria verrucaria	-	pH 5, half-life: 16 min	171652
	Aspergillus cf. niger BCC14405	Q6QA21	30 min, 60% residual activity	665441
	Aspergillus flavus	B8NKE9	purified extracellular enzyme, stable up to	752113
	Aspergillus giganteus	-	xylanase I, half-life 22.5 min, xylanase II, half-life 17.5 min	665016
	Aspergillus niger	-	stable for at least 90 min	664489
		-	purified native enzyme, pH 5.0-7.0, 86% residual activity after 45 min, the crude enzyme retains only 6% of the initial activity	710686
	Aspergillus ochraceus	-	87.5% loss of activity after 90 min	171654
	Aspergillus sulphureus	Q3S401	purified recombinant enzyme, half-life is 30 min	707197
	Aspergillus tubingensis	-	after 1 h 85.9% activity remaining, after 3 h 65.4% activity remaining	707209
	Bacillus amyloliquefaciens	-	stable for several days	171646
	Bacillus licheniformis	-	X-I retains more than 90% of hydrolytic activity after 1 h of heating at 50°C	677623
		B5SYI8	is stable at temperatures up to 50°C. At 55°C, more than half of the activity is lost, and no activity remains at 60°C	697502
		-	partially purified extracellular xylanase, completely stable for 30 min, loss of 50% activity after 120 min	752334
	Bacillus pumilus	-	half-life: 30 min	171586
		-	30 min, 50% loss of activity	171624
	Bacillus sp. (in: firmicutes)	-	pH 6.0, 10 min, enzyme component I from strain W2, stable up to	171595
		Q84F19	3 h, stable	664690
		-	purified native enzyme, stable	708851
	Bacillus subtilis	B9ZZN9	the half-life of the recombinant protein at 50°C is 65 min whereas it is approximately 25 min at 60°C. The enzyme is not stable at 70°C and it loses almost all of its activity even 5 min after incubation	699055
	Cellulomonas uda	-	1 h, 24% loss of activity	136300
	Cellulosimicrobium cellulans	T2M1V1	30 min, 50% residual activity	731029
	Cellulosimicrobium sp.	B2BZ80	stable below for at least 1 h	737466
	Cohnella sp. AR92	-	isoforms Xyn10A and CoXyn10C are efficient at moderate temperatures; however, their half-life time as a function of the activity at 50°C is of 120 min and 60 min with a	768810

		residual activity of 48% and 52%, respectively. Isoform Xyn11A shows a loss of approximately 60% of its activity after 20 min of incubation at 50°C	
Cutaneotrichosporon cutaneum	-	10 min, complete loss of activity	171625
Fusarium commune	A0A0F7MX47, A0A0F7MYK0, A0A0F7N1T1	1 h, more than 40% resiudal activity	738135
Malbranchea pulchella	-	pH 6.7, 30 min, stable	171587
Melanocarpus albomyces	-	stable	738730
	-	pH 5-10, stable for 24 h, both isoforms xylanase IA and xylanase IIIA	738730
Microcella alkaliphila	A0A0U5B5Z4	half-life 48 h	738192
Naganishia albida	-	10 min, 5% loss of activity	171631
	-	pH 5.4, 0.1 M NaCl, half-life: 7 min	171585
Paecilomyces sp. 'thermophila'	D1G4K3	recombinant His-tagged enzyme, 30 min, over 90% activity remaining	707603
Paenibacillus curdlanolyticus	B1A3N2	-	699498
Penicillium occitanis (nom. inval.)	-	purified enzyme, 1 h, 70% activity remaining	718338
Penicillium sclerotiorum	-	half-life of both isozymes is about 80 min	707211
Phanerodontia chrysosporium	-	isoform xynB, stable up to	664683
Pseudopedobacter saltans	F0S6Y3	purified recombinant enzyme, pH 6.0, 90 min, loss of 70% activity	752084
Saccharopolyspora pathumthaniensis	E1APH5	purified enzyme, 2 h, stable	717401
Schizophyllum commune	P35809	5 min, 40% residual activity	729654
Staphylococcus sp.	-	half-life: 240 min	171641
Streptomyces coelicolor	Q8CJQ1	1 h, 90% residual activity	737481
Streptomyces griseorubens	-	1 h, 60% residual activity	745416
Streptomyces lividans	-	pH 6.0, 24 h, about 80% loss of activity	171606
Streptomyces sp.	-	Xyl1-Xyl3 are stable for 16 h	655355
	F7J663	stable up to, for 1 h	730370
Streptomyces sp. TN119	C6FX34	purified recombinant enzyme, 1 h, 95% remaining activity, rapid decrease in activity above 60°C	707196
Talaromyces funiculosus	Q8J0K5	5 min, 25% residual activity	664181
Talaromyces pinophilus	-	isozyme III is stable below	655807
	A0A6N4SL16	the enzyme is stable at temperatures of up to 50°C during 24 h of incubation at pH 4.0	768726
Thermoascus aurantiacus	-	12 weeks, more than 50% residual activity	742421

	Thermoascus aurantiacus var. levisporus	-	90 min, 70% residual activity	743330
	Thermochaetoides thermophila	Q8J1V6	purified recombinant enzyme expressed from gene with intron, half-life is 176 min	709894
		-	stable for 24 h, xylanase I, xylanase II	737964
	Thermomyces dupontii	-	stable for 7 days	741629
		-	pH 7-8, stable for 7 days	741630
	Thermomyces lanuginosus	-	24 h, 50% residual activity	742214
		-	pH 6.5, 51 h, 100% residual activity	742958
		O43097	t1/2: 693 min, fusion enzyme (EG-M-Xyn) of endoglucanase (cellulase) from Teleogryllus emma and xylanase from Thermomyces lanuginosus, xylanase activity	750783
		O43097	t1/2: 173 min, fusion enzyme (EG-M-Xyn) of endoglucanase (cellulase) from Teleogryllus emma and xylanase from Thermomyces lanuginosus, endoglucanase activity	750783
	Thermothelomyces heterothallicus	-	pH 5.0, 24 h, 80% remaining activity	655176
	Trametes hirsuta	-	30 min, stable	171629
	Trichoderma harzianum	-	1 h, 29000 Da enzyme form is stable	171613
	Trichoderma reesei	-	20 h, 27% loss of activity of xylanase I and II	136301
		B2CNY5	purified recombinant enzyme, 30 min, 94% activity remaining	707864
		B2CNY5	over 85% activity remaining after 30 min	709576
		-	purified enzyme, 30 min, loss of less that 10% activity	707911
	uncultured microorganism	E2DQY5	xylan as substrate, remains stable after treatment for 1 h	717134
50 - 60 ⊟	Acetivibrio thermocellus	-	the enzyme activity remains stable after 16 h at 50°C. At 60°C, the enzyme shows less than 40% activity after 3 h	750432
	Rhodothermaceae bacterium RA	A0A1V0DJ74	the enzyme shows a half-life of 1 h at 60Â°C. When the pre-incubation is carried out at 50Â°C, the enzyme retains 80% of its maximal activity for 5 h. The protein is unstable at 70Â°C	775585
	Streptomyces fradiae	A7TVD4	remains stable at 50°C after incubation for 30 min at pH 7.8. More than 40% of the activity is retained after treating the enzyme at 60°C for 5 min	699489
	Trichoderma sp.	-	purified EX2, 48 h, 50% activity remaining	707762
50 - 65 ⊟	Aspergillus fumigatus	-	the enzyme is stable at 50 and 55Â°C, while activity decreases to 60% when incubated at 60Â°C for 6 h and 19% after 15 min at 65Â°C. The enzyme has a melting temperature of 63.3Â°C	769929
	Streptomyces cyaneus	-	the half-lives of xylanase at 50°C and 65°C are approximately 200 and 50 min, respectively	678757
50 - 80	Streptomyces sp. S9	B4XVN1	retains more than 65% of the maximum activity	695801
50 - 85	Priestia megaterium	-	remains stable at 50°C for 20 min and at 85°C for 7 min	681810
52 - 65	Bacillus subtilis	-	the enzyme activity decreases above 52°C showing no residual activity above 65°C when pre-incubated for 2 min	770731

55 ⊟	Acrophialophora nainiana	-	purified enzyme is stable for 44 h	656332
	Aspergillus caespitosus	-	isoform xyl I, half-life 27.3 min, xyl II, half-life above 90 min	666701
	Aspergillus niger	I3QKR9	1 h, mutant enzyme T205C/A52C retains more than 80% of its initial activity	751243
	Bacillus amyloliquefaciens	-	pH 6.8, 15 min, stable up to	171646
	Bacillus sp. (in: firmicutes)	-	pH 5.3, stable for 3 days	171656
		-	30 min, 70% loss of activity, recombinant enzyme	171650
		-	sharp decline in activity above at pH 6.0	654996
	Cellulomonas uda	-	half-life	136300
	Fusarium proliferatum	-	fully stable up to, 30 min, pH 5.0	657201
	Kocuria sp. Mn22	B8XY24	purified recombinant enzyme, 60 min, 70% activity remaining	709503
	Niallia circulans	P09850	the wild type enzyme retains 34% residual activity after 5 min at 55°C, while mutant enzyme R49N retains 43% residual activity after 5 min at 55°C	769727
	Paenibacillus sp.	A0A075EEX4	4 h, 65% residual activity	730187
	Penicillium sclerotiorum	-	half-life of both isozymes is 50 min	707211
	Ruminococcus flavefaciens	-	half-life of the catalytic domain XYLA-A1 is 20 min, half-life of the catalytic domain XYLA-C2 is 50 min	171648
	Staphylococcus sp.	-	half-life: 30 min	171641
	Streptomyces sp.	-	pH 5.5, 30 min, stable	171592, 171594
		-	10 min, stable up to	171619
		-	30 min, stable	171660
	Talaromyces pinophilus	-	isozymes I and II are stable below	655807
	Termitomyces aurantiacus	-	10 days, about 90% residual activity	742400
	Thermomyces lanuginosus	-	30 min, stable up to	171617
		-	10 days, about 90% residual activity	742400
		-	pH 10, stable for 5 h	646829
	Trichoderma reesei	LC132960	30 min, almost complete loss of activity for wild-type, 9% residual activity for mutant Q286A/N340Y	744124
	uncultured microorganism	I6M1A2	2 h, 90% residual activity	731026
55 - 85	Bacillus subtilis	-	30 min, enzyme retains more than 70% of its initial activity	751229
55.2	Teredinibacter turnerae	C5BJ89	melting temperature of apoenzyme	749504
56	Aspergillus nidulans	-	9 h, more than 90% of maximal activity remains	171658
57.3	Teredinibacter turnerae	C5BJ89	apoenzyme with xylan	749504
58.9	Talaromyces pinophilus	W8VR85	wild-type, melting temperature	737727
60 ⊟	Acrophialophora nainiana	-	half-life 2 h	655620

	-	purified enzyme is stable for 1 h	656332
Alicyclobacillus sp.	D3VWB5	purified recombinant XynA4, 1 h, pH 7.0, over 90% activity remaining, half-life is 394 min	717947
Aspergillus flavus	B8NJ86	purified native, inactivation	750988
Aspergillus fumigatus	-	30 min, 40% residual activity	737935
Aspergillus giganteus	-	xylanase I, half-life 1.5 min, xylanase II, half-life 1.5 min	665016
Aspergillus nidulans	-	half-life: 26 min	171658
Aspergillus niger	-	half-life of 115 min	664489
Aspergillus versicolor	-	half-life 1.7 min	666703
Bacillus licheniformis	-	X-II retains more than 90% of hydrolytic activity after 1 h of heating at 60°C	677623
	-	partially purified extracellular xylanase, loss of 10% activity after 60 min, loss of 55% activity after 120 min	752334
Bacillus pumilus	-	complete inactivation in 15 min	171586, 171624
	Q5EFR9	purified recombinant enzyme, 1 h, over 50% activity remaining	717410
Bacillus sp. (in: firmicutes)	-	pH 5.3, 93% loss of activity after 1 day	171656
	-	stable	171645
	-	stable up to	171621
	-	pH 6.0, 10 min, enzyme component II from strain W1 and W2, stable up to	171595
	-	purified enzyme, 5 h, 95% activity remaining	717105
Bacillus subtilis	-	50% final residual activity in presence of 0.5 M trehalose as well as 8fold increased half-life of XylII and 2.5fold of XylI, inactivation in absence of trehalose	655381
	-	the wild type enzyme completely loses activity after preincubation at 60°C	677993
	Q45VU2	glycosylated recombinant enzyme, 30 h, 50% remaining activity. Unglycosylated native enzyme, 12 h, 50% activity remaining	707810
Bispora sp. MEY-1	D6MYS9	purified recombinant enzyme, 60 min, pH 3.0, 98% remaining activity	707258
Cellulomonas uda	-	half-life: 10 min	136300
Demequina sp. JK4	B9VSZ3	purified recombinant enzyme, 80% activity remaining	709491
Fusarium proliferatum	-	loss of 78% activity after 30 min at pH 5.0	657201
Irpex lacteus	-	pH 5.0, 10 min, stable up to	171612
Kocuria sp. Mn22	B8XY24	purified recombinant enzyme, 60 min, 30% activity remaining	709503
Laetiporus sulphureus	-	retains approximately 97% of its relative activity following incubation for 4 h	699053
Luteimicrobium xylanilyticum	A0A286LNP2	1 h, 80% residual activity	745089
Melanocarpus albomyces	-	isozyme IIb shows a half-life of 94.5 min, isozyme IId of 85.6 min, isozyme Ia of 164.3 min	654986

Mycothermus thermophilus	-	45 min, xylanase 2 completely loses activity, xylanase 1 retains 50% of its activity	738127
Parageobacillus thermantarcticus	-	stable for 24 h	657331
Pediococcus acidilactici	-	purified extracellular enzyme, 2 h, loss of 80% activity, half-life is 20 min	750814
Penicillium canescens	C3VEV9	half-life 7 h	729182
Penicillium occitanis (nom. inval.)	-	purified enzyme, 1 h, 50% activity remaining	718338
Phanerodontia chrysosporium	-	isoforms xynA, xynC, stable for 30 min	664683
Pseudopedobacter saltans	F0S6Y3	purified recombinant enzyme, pH 6.0, 90 min, inactivation	752084
Pseudozyma hubeiensis	-	purified PhX20, half-life is 10 min	707766
Saccharopolyspora pathumthaniensis	E1APH5	purified enzyme, half-life is 180 min	717401
Staphylococcus sp.	-	half-life 10 min	171641
Streptomyces coelicolor	Q8CJQ1	1 h, 16% residual activity	737481
Streptomyces sp.	-	half-lives: Xyl1 10 min, Xyl2 30 min, Xyl3 70 min	655355
Talaromyces purpureogenus	A0A2H4RAC5	purified recombinant enzyme, pH 6.5, 60 min, stable	749635
Termitomyces clypeatus	-	stable up to	171630
Thermoascus aurantiacus	-	97 h, retains full activity	171601
	-	retains full activity for 97 h	171601
	-	half-life 4 days	742421
Thermochaetoides thermophila	Q8J1V6	purified recombinant enzyme expressed from gene with intron, half-life is 72 min	709894
	G0S9G3	120 min, enzyme retains about 85% of its initial activity, xylanase ativity	750799
	G0S9G3	120 min, enzyme retains more than 80% of its initial activity, endoglucanase activity	750799
	G0S9G3	purified recombinant His-tagged enzyme, 90% activity remaining after 120 min	750799
Thermoclostridium stercorarium subsp. thermolacticum	-	4 days, 10% loss of activity	171649
Thermomyces lanuginosus	-	stable for more than 48 h	171643
	O43097	96 h, 80% residual activity	731025
	-	stable	737547
	-	4 h, 50% residual activity	742214
	-	pH 5-9, stable for 5 h	646829
	O43097	t1/2: 347 min, fusion enzyme (EG-M-Xyn) of endoglucanase (cellulase) from Teleogryllus emma and	750783

			xylanase from Thermomyces lanuginosus, xylanase activity	
		O43097	t1/2: 16 min, fusion enzyme (EG-M-Xyn) of endoglucanase (cellulase) from Teleogryllus emma and xylanase from Thermomyces lanuginosus, endoglucanase activity	750783
	Thermothelomyces heterothallicus	-	half-life 150 min	655176
	Thermotoga maritima	-	recombinant mutant hybrid enzyme, 30 min, over 85% activity remaining	710126
	Trichoderma reesei	-	20 h, 84% loss of activity of xylanase I, 76% loss of activity of xylanase II	136301
		B2CNY5	over 65% activity remaining after 30 min	709576
		-	purified enzyme, 30 min, loss of about 70% activity	707911
		-	recombinant mutant hybrid enzyme, 30 min, over 85% activity remaining	710126
		LC132960	30 min, 51% residual activity, mutant Q286A/N340Y	744124
	Trichoderma virens	-	10 min, complete loss of activity	171632
	uncultured bacterium	B2LWN3	Xyn11 in its apo form, is not subject to thermal inactivation up to 60°C	698785
	uncultured microorganism	E2DQY5	xylan as substrate, remains 60% activity after treatment for 1 h	717134
		E2DQY5	purified recombinant RuCelA, 1 h, 60% activity remaining	717134
60 - 65	Trichoderma reesei	-	marginally stable at temperatures above 60°C, with half-life of about 0.5 min at 65°C	680210
60 - 70 ⊟	Actinomadura sp.	-	purified recombinant XynS14s, 12 h, completely stable	717886
	Halalkalibacterium halodurans	-	stable up to 60°C, retains 30% of original activity after incubation at 70°C	681342
	[Caldibacillus] cellulovorans	Q9L8L8	thermostability of the recombinant catalytic domain is decreased to 60°C in absence and increased to 70°C in presence of the recombinant N-terminal thermostabilizing domain D1	656718
60 - 80	Streptomyces sp. T7	-	the enzyme shows excellent thermostability, retaining over 90% of the residual activity after 60 min of incubation at 60°C. Furthermore, the enzyme retains more than 40% activity when incubated at 80°C for 60 min	774465
60.1	Teredinibacter turnerae	C5BJ89	apoenzyme with xylohexaose	749504
64.6	Talaromyces pinophilus	W8VR85	mutant D65P/N66G, melting temperature	737727
65 ⊟	Alicyclobacillus sp.	D3VWB5	purified recombinant XynA4, half-life is 17 min	717947
	Aspergillus fumigatus	-	1 h, 50% residual activity	742216
	Geobacillus sp. WBI	B5M201	1 h, pH 10, 100% residual activity	738547
	Geobacillus stearothermophilus	-	pH 7.0, 10 h, stable, pH 9.0, half-life is 6 h	171651
	Malbranchea cinnamomea	-	1 h, 43% residual activity	741623
	Neocallimastix patriciarum	Q9UV68	20 min, wild-type, 70% loss of activity, mutant N38Y/F52W/G56Y/G201L, about 10% loss of activity	744133
	Paenibacillus sp.	-	purified native enzyme, half-life is 415.12 min	717094
	Streptomyces ipomoeae	MK331807	1 h, no loss of activity	750086

	Streptomyces sp. TN119	C6FX34	purified recombinant enzyme, 1 h, 70% remaining activity	707196
	Thermoclostridium stercorarium	-	15 days, stable	171615
	Thermopolyspora flexuosa	-	recombinant His-tagged enzyme, 30 min, completely stable up to	707229
	Thermothelomyces fergusii	-	1 h, 61% residual activity	741623
	Thermothelomyces thermophilus	-	1 h, 90% residual activity	742216
	Thermothielavioides terrestris	-	half-life 23 days	742470
		-	6 h, 80% residual activity	741661
	Trametes hirsuta	-	30 min, 90% loss of activity	171629
	Trichoderma reesei	-	half-lives: wild-type enzyme below 1 min, S110C/N154C 14 min, S110C/N154C/Q162H and S110C/N154C/Q162Y 63 min, S110C/N154C/Q162H/N11D 107 min, S110C/N154C/Q162Y/N11D 97 min, S110C/N154C/Q162H/N11D/N38D 112 min	656334
	Trichoderma viride	-	30 min, complete inactivation	171623
66 ⊟	Aspergillus nidulans	-	half-life: 3 min	171658
	Thermomyces lanuginosus	O43097	melting temperature, wild-type	742096
70 ⊟	Acetivibrio thermocellus	P10478, O32374	2 h, 87% activity of mutant Xyn-BDC is remaining, 42% of mutant Xyn-BDC	709200
		P10478, O32374	2 h, mutant Xyn-C loses all activity, while mutant XynC-BC retains 70% activity	709200
	Alicyclobacillus sp.	D3VWB5	purified recombinant XynA4, half-life is 3 min	717947
	Aspergillus fumigatus	E0X4B3	1 h, complete loss of activity for wild-type, 46% residual activity for mutant T16A/T39I/L176Q, 33% for mutant A60D, 25.8% for mutant S68R, 23.8% for mutant Q47P/S159R	738888
	Aspergillus niger	F5CI28	10 min, complete loss of activity	729014
	Bacillus licheniformis	-	the recombinant enzyme is stable below 70°C, enzyme activity is rapidly lost above 70°C, the residual activity of the recombinant BlxA is 76% at 70°C and pH 6.0 after 2 min	682685
		-	partially purified extracellular xylanase, loss of 80% activity after 120 min	752334
	Bacillus sp. (in: firmicutes)	-	complete inactivation after 1 day	171656
		-	pH 6.0, 4 h, 20% loss of activity in presence of xylan, 80% loss of activity without xylan	171645
		-	15 min, stable	171628
		-	4 h, in presence of xylan, 20% loss of activity	171645
		-	4 h, 20% activity remaining	717091
		-	purified enzyme, 5 h, 82% activity remaining	717105
	Bacillus subtilis	-	35% final residual activity in presence of 0.5 M trehalose as well as 2fold increased half-lives of XylII and XylI, inactivation in absence of trehalose	655381

	Q45VU2	glycosylated recombinant enzyme, 1 h, 90% remaining activity	707810
	-	the thermostable variant XynAG3 loses 90% of its activity after 10 min, and after 30 min no activity is observed. The chimeric CotA-XynAG3 retains 60% activity after 10 min and about 20% activity after 120 min. The half-lives of the chimeric enzymes XynAG3 and CotA-XynAG3 are 2.2 and 21 min	719992
	A0A172MAU1	3 h, immobilized enzyme retains 45.3% of its activity, soluble enzyme retains 16.3% of its activity	750593
Caldicellulosiruptor saccharolyticus	-	pH 6.0, half-life: 4 min in absence of added protein, 30 min, in presence of 0.5 mg/ml bovine serum albumin	171655
Geobacillus stearothermophilus	-	pH 7.0, half-life is 14.5 h	171651
Irpex lacteus	-	pH 5.0, 10 min, inactivated	171612
Kocuria sp. Mn22	B8XY24	purified recombinant enzyme, 60 min, 10% activity remaining	709503
Malbranchea cinnamomea	A0A088BC14	half-life 76 min	742680
Melanocarpus albomyces	-	isozyme Ia shows a half-life of 16.7 min, isozyme IId of 15.6 min	654986
Nesterenkonia xinjiangensis	D1KJJ7	purified enzyme, pH 7.0, 60 min, 55% activity remaining	707216
Paecilomyces sp. 'thermophila'	D1G4K3	recombinant His-tagged enzyme, half-life is 174.8 min	707603
Paenibacillus sp.	-	purified native enzyme, half-life is 102.8 min	717094
Parageobacillus thermantarcticus	-	half-life 24 h	657331
Penicillium janczewskii	-	24 h, more than 70% residual activity	739567
Penicillium occitanis (nom. inval.)	-	purified enzyme, 15 min, 68% activity remaining	718338
Phanerodontia chrysosporium	-	isoforms xynA, xynB, xynC, complete inactivation in 30 min	664683
Phialophora sp. CGMCC 3328	F2VRY7	XYN10G, about 90% of activity remaining after 1 h, 37°C	717131
Pseudozyma hubeiensis	-	purified PhX33, half-life is 15 min	707766
Rasamsonia byssochlamydoides	-	pH 5.5, 5 min, xylanase X-a, X-b-I and X-b-II, stable up to	171627
Rhizophlyctis rosea	A0A2U8LMQ3	1 h, 56% residual activity (small enzyme form RrXyn11A S), 1 h, 65% residual activity (larger enzyme form RrXyn11A L)	749718
Saccharopolyspora pathumthaniensis	E1APH5	purified enzyme, half-life is 120 min	717401
Streptomyces avermitilis	Q9X584	purified heavier protein band (42.8 kDa), retains more than 40% after a 1 h incubation at 70°C. Retains 35% of its activity after a 1 h incubation at 87°C	698373
Streptomyces ipomoeae	MK331807	1 h, 15% loss of activity	750086
Streptomyces sp.	-	30 min, complete inactivation	171660
Streptomyces sp. TN119	C6FX34	purified recombinant enzyme, 1 h, 45% remaining activity	707196

Talaromyces funiculosus	Q5ZNB1	purified recombinant enzyme, 180 min, no loss of activity	718080
Termitomyces clypeatus	-	1 h, complete loss of activity	171630
Thermoascus aurantiacus	-	24 h, retains full activity	171601
	-	retains full activity for 24 h	171601
	-	hlaf-life 1.5 days	742421
	-	half-life 204 h	742959
	-	stable for 8 h	741753
	-	stable up to 6 h	735783
Thermobifida fusca	-	both recombinant TfxA and mutant 2TfxA98 are stable at 70°C during 30 min incubation, although the mutant is slightly less stable than the parent at 70°C after 30 min of incubation	696903
Thermochaetoides thermophila	Q8J1V6	purified recombinant enzyme expressed from gene with intron, half-life is 39.4 min	709894
	G0S9G3	60 min, enzyme retains 66.2% of its initial activity, xylanase ativity	750799
	G0S9G3	60 min, enzyme retains 74.3% of its initial activity, endoglucanase activity	750799
	G0S9G3	purified recombinant His-tagged enzyme, half-life of endoxylanase activity is 100 min, 40% activity remaining after 120 min	750799
Thermoclostridium stercorarium	-	10 min, xylanase A, stable	171582
	-	stable up to	171661
Thermomyces lanuginosus	-	mutant enzyme retains over 50% activity after 2 h	655088
	O43097	80 min, 60% remaining activity, native and recombinant enzyme	654274
	-	30 min, stable up to. 120 min, 67% residual activity	742766
	O43097	wild-type at pH 8.0, half-life 11 min in presence of substrate, 33 min in absence of substrate, respectively. Mutant Q1C/Q24C at pH 8.0, half-life 229 min in presence of substrate, 315 min in absence of substrate, respectively. Mutant Q1C/Q24C at pH 9.0, half-life 112 min in presence of substrate, 169 min in absence of substrate, respectively	742096
	O43097	t1/2: 231 min, fusion enzyme (EG-M-Xyn) of endoglucanase (cellulase) from Teleogryllus emma and xylanase from Thermomyces lanuginosus, xylanase activity	750783
	O43097	t1/2: 15 min, fusion enzyme (EG-M-Xyn) of endoglucanase (cellulase) from Teleogryllus emma and xylanase from Thermomyces lanuginosus, endoglucanase activity	750783
Thermothelomyces heterothallicus	-	half-life 15 min	655176
Thermothelomyces thermophilus	G2QIK8, G2QDB9	purified recombinant enzyme, 30 min, 60% activity remaining	750646
Trichoderma reesei	-	20 h, complete inactivation	136301

		-	10 min: inactivation of mutant S110C/N154C, 50-60% remaining activity of mutants S110C/N154C/Q162H/N11D, S110C/N154C/Q162Y/N11D, and S110C/N154C/Q162H/N11D/N38D	656334
		-	purified enzyme, 30 min, loss of over 90% activity	707911
		-	purified recombinant enzyme, loss of 60% activity after 30 min	707415
		-	the enzymeactivity remains stable after 30 min at 70°C, but decreases at higher temperatures. No residual activity is detected above 80°C	770736
	uncultured microorganism	E2DQY5	xylan as substrate, almost loses its activity after treatment for 1 h	717134
		E2DQY5	purified recombinant RuCelA, 1 h, almost no activity remaining	717134
70 - 75	Rhizomucor miehei	-	purified enzyme, 60 min, completely stable	717022
70 - 90	Halalkalibacterium halodurans	-	stable up to 70°C, 2% and 16% of the original activity are retained after incubation at 80°C and 90°C, respectively	681342
70.4	Thermoascus aurantiacus	-	melting temperature, mutant N257D	738405
70.9	Thermoascus aurantiacus	-	melting temperature, wild-type and mutant Q158R	738405
71.3	Thermoascus aurantiacus	-	melting temperature, mutant H209N	738405
74	Thermomyces lanuginosus	O43097	melting temperature, mutant Q1C/Q24C	742096
74.5	Talaromyces pinophilus	W8VR85	mutant S35C/T62C, melting temperature	737727
75 ⊟	Bacillus amyloliquefaciens	-	pH 6.8, 15 min, nearly all activity is destroyed	171646
	Bacillus sp. (in: firmicutes)	-	15 min, inactivation	171628
	Geobacillus stearothermophilus	-	pH 7.0, half-life is 20 min	171651
	Microcella alkaliphila	A0A0U5B5Z4	30 min, stable up to 75°C	738192
	Paecilomyces sp. 'thermophila'	D1G4K3	recombinant His-tagged enzyme, half-life is 137.9 min	707603
	Pseudothermotoga thermarum	F7YXD6	2 h, 75% resiudal activity	729475
	Streptomyces ipomoeae	MK331807	t1/2: 30 min	750086
	Thermoclostridium stercorarium	-	1 h, xylanase A and C are stable, xylanase B retains 85% of its activity	171615
	Thermomyces lanuginosus	-	30 min, 80% loss of activity	171643
75 - 95	Halalkalibacterium halodurans	-	the enzyme retains more than 80% activity when incubated at 75°C for 2 h, 37% activity when incubated at 85°C for 2 h, and 18% activity after incubation at 95°C for 60 min	774238
80 ⊟	Acetivibrio thermocellus	A3DJS9	melting temperature, Ca2+ ions provide extra thermal stability	773744
	Actinomadura sp.	-	XynS14 expressed from Pichia pastoris shows 50% remaining activity after 2 h, the enzyme expressed from	717886

		Escherichia coli shows 30% remaining activity. In the presence of sorbitol at 90% and glycerol at 50%, the enzyme retains approximately 70% and 80% activity, respectively, at 80°C for 4 h	
Aspergillus niger	-	15 min, xylanase IA and Ib are rapidly inactivated above, xylanase II is completly inactivated	171633
	-	15 min, complete inactivation	171634
	-	half-life of 33 min	664489
Aspergillus tubingensis	-	22% activity remaining	707209
Bacillus licheniformis	-	partially purified extracellular xylanase, loss of 85% activity after 120 min	752334
Bacillus sp. (in: firmicutes)	-	15 min, complete loss of activity	171645
Bacillus subtilis	Q45VU2	glycosylated recombinant enzyme, 1 h, 86% remaining activity	707810
Bispora sp. MEY-1	D6MYS9	purified recombinant enzyme, 5 min, pH 3.0, 25% remaining activity	707258
	D0QF43	purified recombinant Xyl10C, 60 min, completely stablem half-life is 45 h	708455
Caldicellulosiruptor owensensis	E4Q2A4	half-life 1 h	745147
Caldicellulosiruptor saccharolyticus	-	pH 6.0, half-life is 20 min in presence of 0.5 mg/ml bovine serum albumin	171655
Halalkalibacterium halodurans	M4QNR9	wild-type, half-life 35 min, recombinant glycosylated enzyme, half-life 45 min	737851
Melanocarpus albomyces	-	all isozymes are inactivated within 10 min	654986
Melioribacter roseus	I7A603, I6Z823, I6Z9A7	half-life 1 h	738862
Nesterenkonia xinjiangensis	D1KJJ7	purified enzyme, pH 7.0, 60 min, 10% activity remaining	707216
Parageobacillus thermantarcticus	-	hlaf-life 50 min	657331
Penicillium janczewskii	-	half-life 1.7 h	739567
Rhizomucor miehei	-	purified enzyme, 60 min, 80% activity remaining	717022
Saccharopolyspora pathumthaniensis	E1APH5	purified enzyme, inactivation	717401
Talaromyces funiculosus	Q5ZNB1	purified recombinant enzyme, 10 min, loss of 80% activity	718080
Thermoascus aurantiacus	-	half-life: 54 min	171601
	-	half-life 32 min	741753
Thermochaetoides thermophila	Q6UN40	retains 35% enzyme activity. The enzyme shows half lives of 1100, 330, 86, 70, 19.3 and 11.4 min at 30, 40, 50, 60, 70 and 80°C, respectively	701091
	Q8J1V6	purified recombinant enzyme expressed from gene with intron, half-life is 18.7 min	709894
	G0S9G3	60 min, enzyme retains about 50% of its initial activity, xylanase ativity	750799

		G0S9G3	60 min, enzyme retains 61.3% of its initial activity, endoglucanase activity	750799
		G0S9G3	purified recombinant His-tagged enzyme, half-life of endoxylanase activity is 40 min, 10% activity remaining after 80 min, inactivation after 120 min	750799
	Thermoclostridium stercorarium	-	enzyme begins to denature at around 80°C	171657
	Thermomonospora sp.	-	half-life 86 min	654413
	Thermomyces dupontii	-	half-life 4 h	741630
	Thermomyces lanuginosus	O43097	40 min, 60% remaining activity, native and recombinant enzyme	654274
		-	loses 90% activity after 10 min at 80°C	680970
		O43097	t1/2: 63 min, fusion enzyme (EG-M-Xyn) of endoglucanase (cellulase) from Teleogryllus emma and xylanase from Thermomyces lanuginosus, xylanase activity	750783
		O43097	t1/2: 12 min, fusion enzyme (EG-M-Xyn) of endoglucanase (cellulase) from Teleogryllus emma and xylanase from Thermomyces lanuginosus, endoglucanase activity	750783
	Thermopolyspora flexuosa	-	recombinant His-tagged enzyme, 30 min, 50% activity remaining	707229
	Thermothelomyces heterothallicus	-	24% remaining activity	655176
	Thermotoga petrophila	A5IL00	8 h, enzyme remains stable	748630
80.5 ⊟	Talaromyces pinophilus	T2HRP0	melting temperature	730881
		W8VR85	mutant S35C/N44H/Y61M/T62C/N63L/D65P/N66G/T101P/S102N, melting temperature	737727
81	Thermoclostridium stercorarium	-	half-life of xylanase A is 90 min, half-life of xylanase B is 2 min 30 s, half-life of xylanase C is 8 min	171615
85 ⊟	Bispora sp. MEY-1	D0QF43	purified recombinant Xyl10C, half-life is 3 h	708455
	Paecilomyces sp. 'thermophila'	D1G4K3	recombinant His-tagged enzyme, 30 min, over 60% activity remaining, half-life is 68.2 min	707603
	Rhizomucor miehei	-	purified enzyme, 60 min, 40% activity remaining	717022
	[Caldibacillus] cellulovorans	Q9L8L8	12% remaining activity of recombinant catalytic domain	656718
90 ⊟	Acidothermus cellulolyticus	-	half-life is 1.5 h in presence of xylan	717123
	Aspergillus niger	-	pH 5.0, 10 min, 40% residual activity	731024
	Bacillus licheniformis	-	X-I presents 50% activity at 90°C, and X-II presents 80% activity at the same temperature	677623
		-	partially purified extracellular xylanase, loss of 90% activity after 120 min	752334
	Bispora sp. MEY-1	D0QF43	purified recombinant Xyl10C, 10 min, 87% remaining activity	708455
	Halalkalibacterium halodurans	-	half-life 60 h	729408
	Nesterenkonia xinjiangensis	D1KJJ7	purified enzyme, pH 7.0, 60 min, 5% activity remaining	707216
	Rhizomucor miehei	-	purified enzyme, half-life is 30 min, 20% activity remaining after 60 min	717022

	Saccharolobus solfataricus	-	50% maximal activity is measured after 85 min at 90°C, 23% residual activity after 3 h at 90°C	679721
	Teredinibacter turnerae	C5BJ89	purified recombinant enzyme, 20 min, inactivation	749504
	Thermobifida fusca	Q47QL8	30 min, more than 10% residual activity, recombinant protein, both with and without His-tag	737932
	Thermochaetoides thermophila	G0S9G3	60 min, enzyme retains about 30% of its initial activity, xylanase ativity	750799
		G0S9G3	40 min, enzyme retains about 28% of its initial activity, endoglucanase activity	750799
		G0S9G3	purified recombinant His-tagged enzyme, half-life of endoxylanase activity is 30 min, inactivation after 80 min	750799
	Thermomonospora sp.	-	half-life 30 min	654413
	Thermomyces dupontii	-	half-life 2 h	741630
	Thermomyces lanuginosus	-	30 min, complete inactivation	171643
		O43097	t1/2: 20 min, fusion enzyme (EG-M-Xyn) of endoglucanase (cellulase) from Teleogryllus emma and xylanase from Thermomyces lanuginosus, xylanase activity	750783
		O43097	t1/2: 12 min, fusion enzyme (EG-M-Xyn) of endoglucanase (cellulase) from Teleogryllus emma and xylanase from Thermomyces lanuginosus, endoglucanase activity	750783
	Thermotoga maritima	-	stable below	663571
95 ⊟	Rasamsonia byssochlamydoides	-	5 min, enzyme form X-a retains 65% of the original activity, enzyme form X-b-I retains 65% of the original activity, enzyme form X-b-II retains 65% of the original activity	171627
		-	t1/2: 5 min, xylanase a	738999
	Saccharolobus solfataricus	Q97YG7	half-life: 53 min	692258
	Thermopolyspora flexuosa	-	recombinant His-tagged enzyme, 30 min, 10% activity remaining	707229
	[Caldibacillus] cellulovorans	Q9L8L8	60% remaining activity of recombinant N-terminal + catalytic domains	656718
96	Thermotoga petrophila	A5IL00	half-life: 54.5 min	748630
additional information ⊟	Aspergillus cf. niger BCC14405	Q6QA21	thermostablitiy increases by 30-50% when 2 M sorbitol is added	665441
		-	thermostablitiy increases by 30-50% when 2 M sorbitol is added	665441
	Aspergillus ochraceus	-	K+ stabilizes against thermal inactivation	171654
	Bacillus subtilis	-	trehalose stabilizes both isozymes at 60°C and 70°C and increases the enzyme activity, thermal deactivation kinetics of enzyme forms XylI and XylII	655381
	Euphausia superba	-	very low thermostability of isozymes A and B	655257
	Melanocarpus albomyces	-	half-lives of the 7 isozymes at different temperatures, isozyme Ia is the most stable, overview	654986
	Niallia circulans	P09850	Tm of wild-type and mutant enzymes	709203
		P09850	thermal denaturation measurements, circular dichroism, overview	707522
		-	thermal denaturation measurements, circular dichroism, overview	707522

	Paenibacillus terrae	F1KBQ3	the absence of the xylan-binding domain results in a decrement of 40% in thermostability	750448
	Pseudoalteromonas haloplanktis	Q8RJN8	psychrophilic enzyme with high activity at low temperatures and low thermal stability	656075
	Thermopolyspora flexuosa	-	the C-terminal carbohydrate binding module has only a slight effect, whereas a polyhistidine tag increases the thermostability of XYN10A	707229
	Thermotoga maritima	-	features that account for the high thermostability of enzyme are clusters of aromatic residues together with a lack of exposed hydrophobic residues	666878